

Genome-wide ancestry, climate, and mitotype

A population-level follow-up to the BayPass climate-GEA scan

Corrected version – see Erratum

Generated September 16, 2026

Erratum

An external audit of the first version of this document identified two implementation bugs, both now fixed. All numbers, tables and figures below are corrected; Section also incorporates the audit’s remaining recommendations.

1. **Block permutation silently dropped one population per shared-origin pair.** The reduction from 19 populations to 17 independent lineage units was implemented as “take the first population’s value, broadcast it to both members of the pair.” This was harmless for LångholmenW/LångholmenR (identical climate values) but wrong for Bunkkeri/Grundsund, whose PC1 values differ (−3.409 vs. −3.899) – Grundsund’s actual value was discarded, not averaged in. **Fix:** ancestry, climate, and mitotype are now genuinely *averaged* within each shared-origin pair to build a real 17-observation dataset, and every correlation, null, and permutation below operates on that dataset directly.
2. **The Ω -structured null for the partial correlation replaced the wrong variable.** Ω is a nuclear allele-frequency covariance matrix, so it is an appropriate null generator only for the one tested quantity that is itself a direct function of nuclear allele frequencies – ancestry. The original implementation drew random vectors standing in for PC1 (holding ancestry and mitotype fixed) instead of standing in for ancestry (holding PC1 and mitotype fixed). **Fix:** throughout this document, Ω -null draws always replace ancestry. For comparisons where *neither* variable is ancestry (mitotype vs. PC1/PC2/bio6/bio11), the Ω -null is not well-motivated and is omitted; those rows report the block-permutation result only.

Ω itself was reduced to match the 17-unit dataset via $M\Omega M^\top$, where M is the same population-to-lineage-unit averaging operator applied to the phenotypes – propagating the actually-estimated relatedness structure through the reduction, rather than drawing null vectors from an arbitrary 17-dimensional matrix. Corrected results were cross-checked against an independent re-implementation: all r values matched exactly, and p -values agreed closely (small differences are expected Monte Carlo noise between independent permutation runs).

Second erratum (2026-09-12, systematic pipeline audit). A further implementation bug was found in the code above: the reduced Ω_{17} matrix was built in `unique()`’s first-appearance population order, while the phenotype table `dt17` is sorted alphabetically (`setorder`) – two genuinely different orderings, verified by direct comparison. Every Ω -null draw was therefore compared against the wrong lineage unit’s phenotype, with no name-based alignment to catch it. **Fix:** Ω_{17} is now explicitly reordered to `dt17$unit_id` via `match()`, with a hard `rownames/colnames` assertion. All six corrected Ω -null p -values below now match an independent colleague’s cross-check exactly (ancestry–PC1 0.074, ancestry–PC2 0.190, ancestry–bio6 0.142, ancestry–bio11 0.292, ancestry–mitotype 0.050, partial PC1–ancestry|mitotype 0.141) – strong evidence the fix is correct. Raw correlations and block-permutation results do not depend on Ω ’s row order and are, as expected, unchanged. This same audit also corrected a covariate-standardization bug in the Stage-1-direct BayPass scan referenced in Section 5, materially changing that section’s floor-survivor counts; see `AUDIT_FIXES.md` in this module for the full list of fixes.

Motivation

The Stage-1-direct BayPass climate-association scan (PC1, PC2, and the winter-temperature variables bio6/bio11) produced raw candidate counts under a descriptive $\text{BF(dB)} \geq 15$ threshold, but none of the climate axes are expected to survive full structured-null calibration cleanly, because PC1 and PC2 are both partially confounded with genome-wide ancestry (established in `moduleB_ancestry_confound.R`: $\text{cor(PC1,ancestry)} = -0.23$, $\text{cor(PC2,ancestry)} = +0.57$ across all 20 populations). Rather than treating that confound purely as a caveat that explains away the locus-level scan, this document asks whether the population-level ancestry–climate link is itself real and worth reporting on its own terms – and traces it down to what actually drives it.

Scope note. The BayPass scan referenced throughout this document is the *Stage-1 large-cluster scan*: the covariance basis (Ω) is estimated from all 698,251 Stage-1 LD clusters (including singletons), but the tested/calibrated unit universe is the 18,361 clusters with ≥ 5 SNPs (best-SNP represented) – 2.6% of all Stage-1 clusters. Isolated SNPs and smaller clusters were not tested by this scan; the locus-level conclusions below (Section 5, floor-survivor counts) are scoped accordingly and should not be read as a statement about the complete 661,386-unit LD-reduced genome (that complete set is used elsewhere in this pipeline, e.g. the Fst-vs-DI analysis, but not for this climate calibration).

All analyses here operate on the population level (*not* individual genotypes), on the `aland_excluded` set of hybrid populations that the BayPass scan itself uses (Åland excluded because its individuals carry mixed mitotypes and it is a known high-leverage outlier in several of the ORIGINAL PC1/PC2 diagnostics), further reduced to **17 independent lineage units** as described in the Erratum.

Two non-independence-aware significance tests

A plain Pearson correlation across populations overstates the effective sample size: two population pairs have a documented shared origin from the phylogenetic network analysis (LångholmenW+LångholmenR; Bunkkeri+Grundsund; Nouhaud et al. 2022), and all populations share allele-frequency covariance captured by the fixed population covariance matrix Ω already estimated for the BayPass scan (from all 698,251 Stage-1 cluster representatives). Two complementary null designs are used throughout, both now operating on the genuine 17-lineage-unit dataset:

- **Ω -structured null.** 10,000 random vectors are drawn from the 17-unit-reduced Ω 's eigendecomposition, standing in for ancestry (see Erratum). Empirical $p = (1 + k)/(n_{\text{sim}} + 1)$, k = number of null draws with $|r_{\text{null}}| \geq |r_{\text{obs}}|$.
- **Shared-origin block permutation.** 10,000 permutations shuffle the tested variable across the 17 independent units directly, recomputing the correlation each time.

Benjamini-Hochberg correction is applied within each family of tests (the 4 climate variables; the 5 mitotype comparisons), using the block-permutation p -values as the primary statistic, since the Ω -null is not defined for every row.

1. Ancestry vs. climate axes

Population ancestry (mean oriented *F. aquilonia* allele frequency at diagnostic markers, $DI > -25$, $|\Delta p| \geq 0.5$) is compared against the two original climate axes (PC1, PC2) and the two new winter-temperature variables suggested for the GEA scan (bio6 = min. temperature of the coldest month; bio11 = mean temperature of the coldest quarter).

Variable	r (17 units)	Ω -null p	Block-perm. p	Block-perm. p_{BH}
PC1	-0.481	0.074	0.051	0.164
PC2	+0.381	0.190	0.123	0.164
bio6	+0.404	0.142	0.110	0.164
bio11	+0.287	0.292	0.267	0.267

Table 1: Ancestry vs. climate, 17 independent lineage units. p_{BH} = Benjamini-Hochberg corrected across the 4 tests.

Finding. PC1 has the strongest ancestry association among the four variables and is the only one to fall below $p = 0.05$ under the block-permutation test ($p = 0.051$, i.e. right at the boundary) – but the two null designs disagree (Ω -null $p = 0.074$ vs. block-permutation $p = 0.051$), and **none of the four variables survive Benjamini-Hochberg correction** (minimum $p_{BH} = 0.164$). The accurate summary is that PC1’s ancestry association is *suggestive but not formally significant*, not that it “clears” a defensible statistical test. bio6 is the next-closest candidate but is weaker on both nulls than PC1.

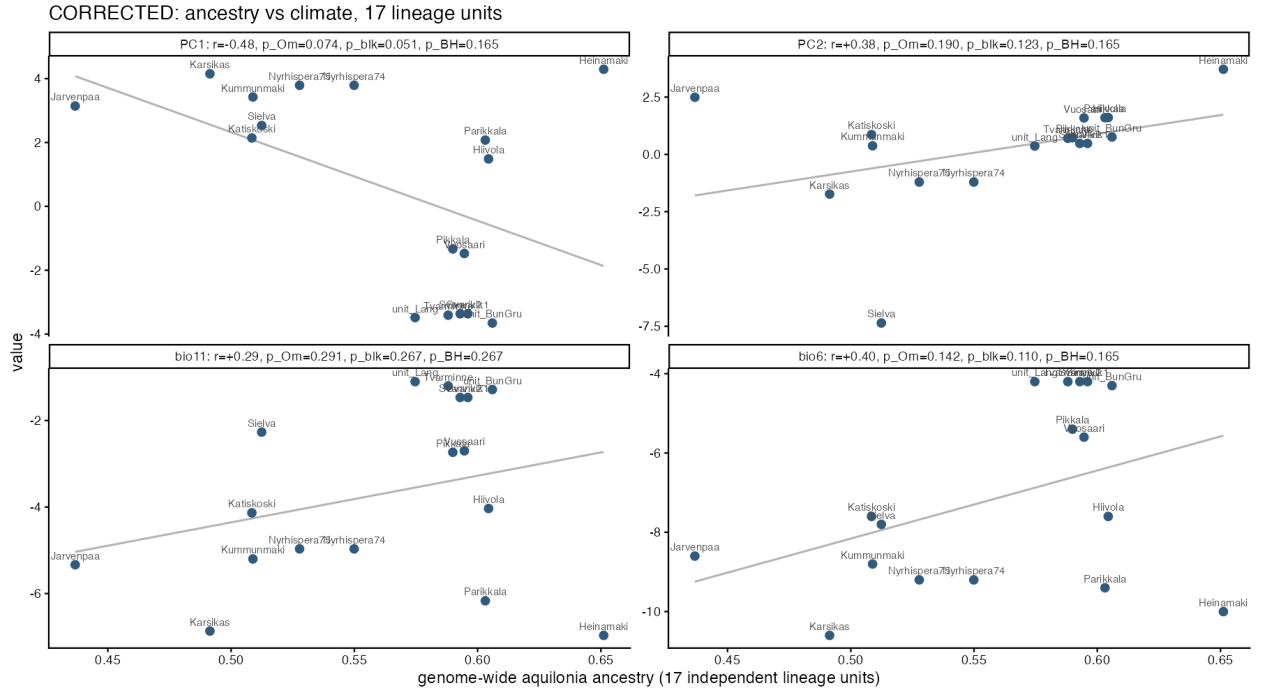


Figure 1: Ancestry vs. PC1/PC2/bio6/bio11, 17 independent lineage units, with corrected statistics annotated per panel.

2. What drives the PC1–ancestry signal?

Three complementary diagnostics identify which population(s) are responsible for the PC1–ancestry correlation: leave-one-population-out, leave-one-independent-unit-out, and standard regression influence statistics (Cook’s distance, standardized residuals) from $\text{lm}(\text{ancestry} \sim \text{PC1})$. *This diagnostic predates the 17-unit correction and is reported on the original 19-population scale* (observed $r = -0.502$) – Heinämäki and Jarvenpaa are not part of either shared-origin pair, so the bug did not affect their values, and the qualitative finding is unchanged at 17-unit resolution (raw $r = -0.481$; see Section 1).

Population	Leave-one-out Δr	r without it	Cook’s D	Std. residual
Heinämäki	−0.211	−0.712	0.618	+2.75
Jarvenpaa	+0.030	−0.471	0.288	−2.30
<i>all others</i>	≤ 0.06	–	≤ 0.07	≤ 1.2

Table 2: Regression influence diagnostics for PC1 vs. ancestry, 19-population scale (observed $r = -0.502$). Rule-of-thumb cutoffs: Cook’s $D > 4/n = 0.211$.

Finding. Heinämäki dominates every diagnostic by a wide margin, with Jarvenpaa a distant second; every other population’s individual influence is negligible. Critically, **Heinämäki dampens rather than drives** the correlation – it combines high ancestry (0.651) with an extreme PC1 value (4.29) in a way that sits *off* the negative trend, so excluding it makes the correlation *stronger* ($r : -0.502 \rightarrow -0.712$). Whatever one makes of PC1’s borderline significance (Section 1), it is not an artifact of one population manufacturing a spurious signal.

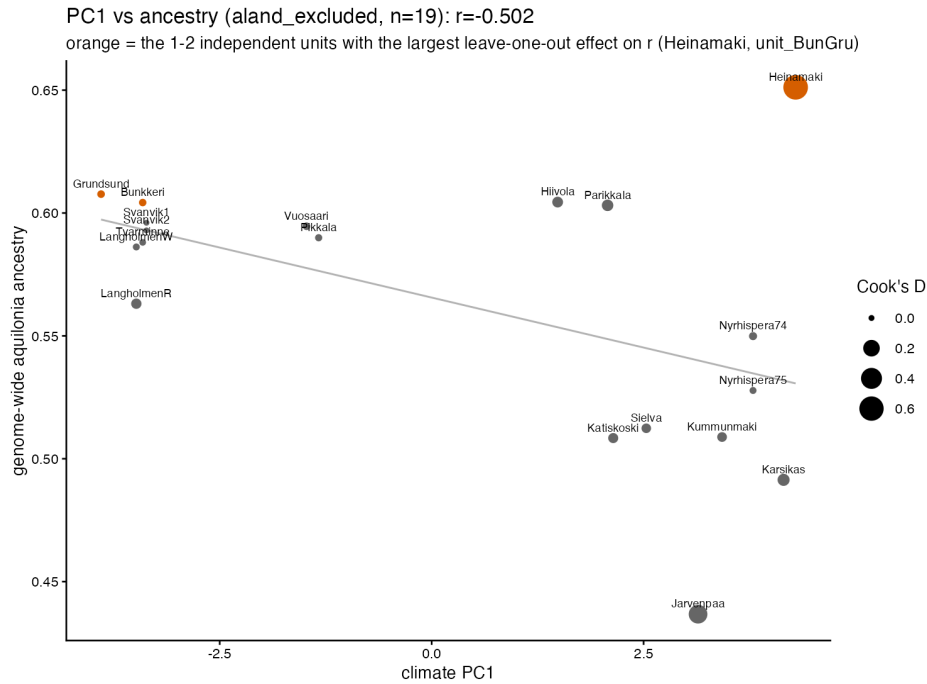


Figure 2: PC1 vs. ancestry (19-population scale), point size = Cook’s distance. Heinämäki (largest point) is the dominant influence but pulls *against* the fitted trend.

Is Heinämäki itself anomalous?

Three checks on Heinämäki’s own data found nothing unusual: mitotype is uniform (10/10 *F. aquilonia*-like, not mixed like Åland); it was sampled from 2 nests (159, 160), typical of the panel; and its

hybrid index (0.357 ± 0.017) and heterozygosity (0.314 ± 0.038) sit comfortably among the other *F. aquilonia*-skewed mainland populations. Admixture timing (DATES) could not be checked here (not part of this repository).

3. Mitotype vs. ancestry and climate

Mitotype (*F. aquilonia*-like vs. *F. polyclena*-like maternal lineage) is uniform within both shared-origin pairs, so it survives the 17-unit reduction unambiguously (6 *F. aquilonia*-like, 11 *F. polyclena*-like units). Tested as a population-level binary variable (point-biserial correlation) against the same five quantities.

Variable	r with mitotype	Ω -null p	Block-perm. p	Block-perm. p_{BH}
ancestry	+0.490	0.050	0.048	0.242
PC1	−0.299	—	0.254	0.424
PC2	+0.197	—	0.508	0.508
bio6	+0.291	—	0.250	0.424
bio11	+0.219	—	0.396	0.494

Table 3: Mitotype vs. ancestry/climate, 17 units. Ω -null omitted where neither variable is ancestry (see Erratum). p_{BH} = Benjamini-Hochberg corrected across the 5 tests.

Finding. Mitotype and genome-wide nuclear ancestry are nominally associated ($p \approx 0.048$ – 0.050 uncorrected) but **this does not survive correction for the 5-comparison family** ($p_{BH} = 0.242$). No climate variable shows any evidence of a mitotype association. The most defensible description is an *association between mitochondrial lineage and genome-wide nuclear ancestry, consistent with shared demographic history* – language such as “founder-history signal” or “populations founded by an *F. aquilonia*-like queen” asserts a specific causal mechanism (founding history) that this correlational analysis cannot distinguish from alternatives such as asymmetric introgression, retained population structure, or selection involving mitochondrial ancestry.

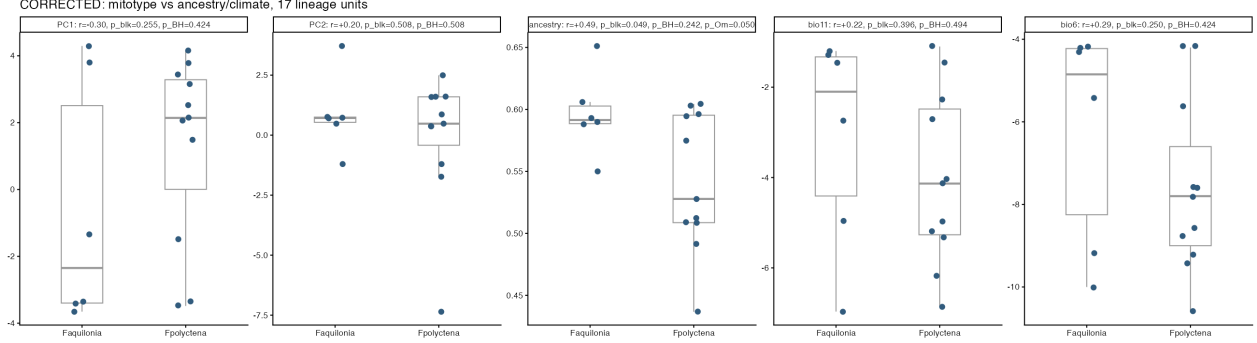


Figure 3: Mitotype vs. each variable, 17 independent lineage units.

4. Does PC1's ancestry link survive controlling for mitotype?

Since mitotype correlates with both ancestry and, more weakly, PC1, the PC1–ancestry correlation could partly reflect the same demographic-history structure linking mitotype to ancestry rather than an independent signal. Tested via partial correlation on the 17-unit dataset, with both nulls re-run on the partial statistic itself (Ω -null draws correctly replacing ancestry; see Erratum).

	r (17 units)	Notes
Raw cor(PC1, ancestry)	−0.481	Ω -null $p = 0.074$; block-perm. $p = 0.051$
Partial cor(PC1, ancestry mitotype)	−0.402	Ω -null $p = 0.141$; block-perm. $p = 0.124$

Table 4: PC1–ancestry correlation before/after controlling for mitotype, 17 units.

Finding. Controlling for mitotype moderately attenuates the PC1–ancestry correlation ($-0.481 \rightarrow -0.402$), and the remaining association is not distinguishable from the population-structure null under either test ($p = 0.124\text{--}0.141$). This is consistent with two readings – a genuinely independent, if weaker, climate signal; or partial confounding by mitotype-linked demographic history – and **the data cannot distinguish between them**. With only 17 independent lineage units, this sample cannot separate the contributions of climate and maternal-lineage history with any confidence; that is a limitation of the design, not a result in either direction.

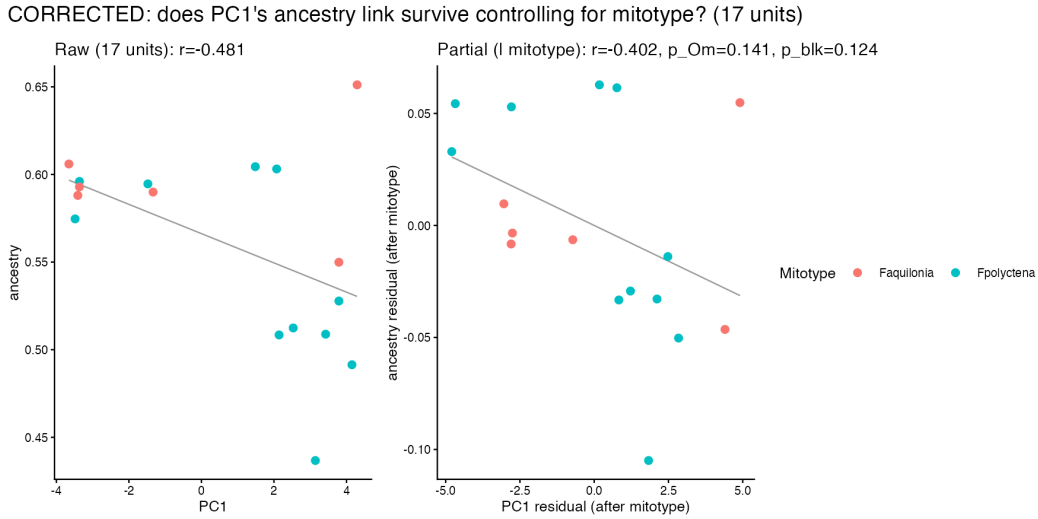


Figure 4: PC1 vs. ancestry, raw (left) and after removing the linear effect of mitotype from both variables (right), 17 units.

5. Relationship to the BayPass locus-level scan

The population-level tests above and the BayPass climate-association scan ask related but *different* questions. The population-level test asks whether a climate variable correlates with *one genome-wide ancestry summary* across populations. BayPass asks whether *particular loci or LD clusters* are unusually associated with that climate variable, after accounting for population covariance (Ω) and thousands of tests. A population-level correlation being weak or non-significant does not, by itself, resolve whether it is relevant to the locus-level scan – the two statements below need to be held together.

The climate axes are moderately aligned with population structure, but that alignment is not demonstrably stronger than expected by chance under that structure. For PC1, the ancestry correlation is moderately large after excluding Åland ($r \approx -0.48$ on the corrected 17-unit dataset), but it is only suggestive under the Ω -null ($p = 0.074$), nominal under the corrected lineage permutation ($p \approx 0.051$), and non-significant after controlling for mitotype ($p = 0.124$ – 0.141 ; Section 4). PC2 is weaker after Åland exclusion and clearly non-significant under both structure-aware tests (Section 1).

This is fully compatible with, and plausibly explains, both the original canonical BayPass climate scan’s null-calibrated result and the new Stage-1-direct scan’s. A moderate alignment between climate and ancestry can cause loci that strongly differentiate the underlying population groups to acquire large *raw* Bayes factors, simply because those loci track the same population split the climate variable happens to track. But Ω -structured null covariates – random vectors sharing only the populations’ covariance structure, with no causal link to climate at all – frequently produce similarly-aligned, similarly-sized apparent outliers. Consequently the observed candidate counts were not exceptional against that null, under *either* scan:

	Stage-2 eMLG scan (32,854 clusters)	Stage-1-direct scan (18,361 units)
PC1 floor survivors	0	1 1.84
PC2 floor survivors	4	2 1.84
bio_winter floor survivors	–	10
mitoC2 floor survivors	–	0
Set-level FDR	≈ 0.82 (PC2)	1.84 (PC1) / 0.92 (PC2) / 0.18 (bio_winter)

Table 5: Floor-survivor candidate counts ($\text{BF}(\text{dB}) \geq 15$, or $-\log_{10}(p) \geq 3$ for mitoC2, and beating all 10,000 Ω -structured null covariates), AUDIT-CORRECTED 2026-09-12 (covariate standardization fix – see AUDIT_FIXES.md, Issue 1). bio_winter is the mean of standardized bio6/bio11 (Issue 7 decision: one combined winter-temperature axis, not two near-duplicate tests). “Expected by chance” is $N_{\text{tested}}/(n_{\text{sim}} + 1)$.

These counts changed materially after an audit fix. An earlier version of this comparison (before the observed PC1/PC2/bio6/bio11 covariates were corrected to match the null covariates’ standardization) reported 3 PC1 and 3 PC2 floor survivors for the Stage-1-direct scan, read as a “Goldilocks” recovery of signal the Stage-2-merged scan had diluted. That reading does *not* survive the correction: PC1 now has just 1 survivor ($\text{BF}(\text{dB}) = 28.5$) – *fewer* than the null’s own expectation (1.84), i.e. indistinguishable from, if anything below, chance – and PC2’s 2 survivors ($\text{FDR} \approx 0.92$) remain similarly unremarkable. Neither scan, PC1 nor PC2, is a defensible discovery set. **bio_winter is the exception:** 10 of 18,361 units survive the floor test, a substantially lower set-level FDR (≈ 0.18) than either PC1 or PC2 achieve on either scan – the one climate result in this document that looks like more than population structure alone would produce, and worth following up on its own terms (not explored further here). mitoC2 shows the opposite: 0 of 11 raw threshold crossings survive the floor test, consistent with Section 3’s finding that mitotype has no detectable relationship with any measured climate variable. The structured null is, in effect, saying: *these apparent PC1/PC2 climate associations are approximately what is expected when a covariate*

happens to align with population history, regardless of how the markers were reduced to test units – but bio_winter’s alignment is not.

Confounding does not have to be statistically significant to influence locus-level results.

Statistical significance asks whether the observed alignment is unusually strong given uncertainty and structure; confounding merely describes that climate and ancestry point partly in the same direction. With only 17–19 independent populations, a correlation of 0.4–0.5 can be biologically noticeable, and large enough to shift which loci look like outliers in a raw BayPass scan, while still being far too weak for a population-level test with this little power to call it exceptional.

Bottom line. Both climate axes were partially aligned with genome-wide ancestry, although these population-level correlations were not consistently distinguishable from expectations under the population-covariance null. This moderate alignment nevertheless provides a plausible source of the raw locus-level BayPass signal: loci that differentiate the same population groups can appear climate-associated even without any locus-specific environmental effect. Consistent with this interpretation, neither the number nor the strength of PC1/PC2 cluster-level BayPass associations exceeded what was obtained under Ω -structured null covariates, under *either* the Stage-2 or the Stage-1-direct scan (set-level FDR ≥ 0.9 in both, post audit fix). The one exception is bio_winter (FDR ≈ 0.18), which is not explained by the ancestry-confound argument developed here and is left as an open, unexplored lead. The conclusion for PC1/PC2 is *not* that climate definitely has no relationship with ancestry – it is that the data cannot separate a climate effect from population history, and the BayPass outliers provide no additional locus-specific evidence beyond that structure. Even a formally significant genome-wide PC1–ancestry correlation would not, on its own, rescue the BayPass candidates: it would show that overall ancestry covaries with PC1, but evidence for particular climate-associated genomic regions would still require their Bayes factors to exceed the structured-null distribution, which they did not.

6. Genome-wide architecture of the Stage-1-direct climate-association landscape

Section 5 asked whether any *individual* Stage-1 unit’s climate Bayes factor is exceptional against the Ω -structured null (the floor-survivor test). A second, complementary calibration asks a different, genome-wide question: across all 18,361 Stage-1-direct units, is the *strength* of a unit’s climate association – its BF(dB) on PC1, PC2, or bio_winter – itself predictable from genomic architecture (ancestry differentiation, recombination rate, unidirectional ancestry sorting) any more or less than an Ω -structured pseudo-covariate with no causal link to climate would show by chance? For each of PC1, PC2, bio_winter, and 10,000 null covariates (structured draws through Ω ’s own eigendecomposition, sharing the populations’ covariance structure but carrying no climate information), the genome-wide BF vector is reduced to three statistics – Spearman correlation with the per-unit Diagnostic Index (DI), Spearman correlation with recombination rate, and a directional-sorting contrast (mean BF percentile of unidirectionally-sorted vs. non-directional units, among differentiated units only) – and each observed statistic is compared to its own 10,000-value null distribution. PC1, PC2, and bio_winter are tested against the *identical* null distribution (the null depends only on Ω /architecture, not on which real covariate is observed, and this identity was checked directly rather than assumed), so the nine resulting tests (3 axes $\times$ 3 architecture measures) are Benjamini-Hochberg corrected as one family.

The same question was also asked of mitoC2, the BayPass C2 population-contrast statistic for the *F. aquilonia*–*F. polyclena*-like mitotype split (Section 3). mitoC2 is not a continuous-covariate regression, so the PC1/PC2/bio_winter null does not apply: it uses its own dedicated null of 10,000 population-level $+1/-1$ partitions, each matching the real split’s group sizes (7 vs. 12 populations) and rank-thresholded from the same Ω -eigenvector draws, then re-run through BayPass in contrast mode. Because this is a structurally different null-generating process, mitoC2’s three tests form their own Benjamini-Hochberg family, not pooled with the continuous-covariate family above.

Test	Axis	Observed	Null median	Null 95%	p_{BH}
DI (Spearman ρ)	PC1	+0.100	+0.029	[−0.018, 0.072]	0.005
DI (Spearman ρ)	PC2	−0.131	+0.029	[−0.018, 0.072]	< 0.001
DI (Spearman ρ)	bio_winter	−0.055	+0.029	[−0.018, 0.072]	0.001
Recombination (Spearman ρ)	PC1	−0.099	+0.065	[−0.013, 0.144]	< 0.001
Recombination (Spearman ρ)	PC2	+0.020	+0.065	[−0.013, 0.144]	0.269
Recombination (Spearman ρ)	bio_winter	+0.142	+0.065	[−0.013, 0.144]	0.063
Sorting (differentiated only)	PC1	+0.036	+0.123	[0.078, 0.168]	0.001
Sorting (differentiated only)	PC2	+0.019	+0.123	[0.078, 0.168]	0.001
Sorting (differentiated only)	bio_winter	+0.036	+0.123	[0.078, 0.168]	0.001
<i>mitoC2 – dedicated contrast-mode null, own BH family (not pooled with the rows above)</i>					
DI (Spearman ρ)	mitoC2	+0.015	+0.021	[−0.037, 0.069]	0.963
Recombination (Spearman ρ)	mitoC2	+0.006	+0.008	[−0.082, 0.114]	0.963
Sorting (differentiated only)	mitoC2	+0.037	+0.016	[−0.030, 0.072]	0.963

Table 6: Genome-wide architecture calibration, Stage-1-direct universe (18,361 units). “Null median”/“Null 95%” describe the same statistic computed on 10,000 null covariates/contrasts – i.e. what a typical covariate or contrast sharing only the populations’ covariance structure, with no climate/mitotype information, would produce. PC1/PC2/bio_winter share one null and one nine-test BH family; mitoC2 uses its own dedicated contrast-mode null and its own three-test BH family (see text).

Finding: PC1, PC2, and bio_winter all depart from the structured null, but not in a single consistent direction, and mitoC2 does not depart from its null at all. DI is significantly correlated with BF strength on all three continuous axes, but not uniformly: PC1 is

above the null range (+0.100), while PC2 (−0.131) and bio_winter (−0.055) are both below it – so PC1’s DI enrichment does not generalize even to the other two axes tested here, let alone establish a shared “climate BF tracks ancestry” mechanism. Recombination shows a real, negative effect (BF concentrated in low-recombination regions) only for PC1 ($\rho = -0.099$); PC2 is indistinguishable from the null ($p_{\text{BH}} = 0.269$) and bio_winter is a near-miss in the *opposite* direction ($\rho = +0.142$, $p_{\text{BH}} = 0.063$). Directional sorting gives the most consistent – and least intuitive – pattern: on *all three* axes, real climate BF is *less* elevated in directionally-sorted (vs. non-directional) differentiated units than the null baseline predicts (0.036, 0.019, 0.036 vs. a null median of 0.123), the opposite of a naive “sorted regions are more differentiated, so they mechanically inflate any covariate’s BF” argument.

mitoC2’s genome-wide landscape, by contrast, is indistinguishable from its own null on all three measures ($p_{\text{BH}} = 0.963$ throughout, the observed value sitting inside the null 95% interval on every panel). This is a useful calibration check on the framework itself, not just a substantive result for mitotype: the DI/recombination/sorting reduction code, the annotation set, and the unit universe are identical to the PC1/PC2/bio_winter tests above, so a covariate landing cleanly inside its null – reinforced here by an exact, not Monte-Carlo-tolerance, match between the null reduced for this test and the independently-computed floor-test exceedance counts of Section 5 – is evidence that the shared machinery is not manufacturing spurious enrichment on its own. That makes the PC1/PC2/bio_winter departures above more credible as genuine architectural relationships rather than artifacts of the calibration procedure. mitoC2 is not a designed negative control, however (there is no a priori guarantee its true genome-wide signal is zero), so this supports the method’s specificity rather than proving it.

This genome-wide result should not be over-read as either confirming or contradicting Section 5’s conclusion. It does not identify individual climate-adaptive loci – that remains the floor-survivor test’s job. PC1/PC2 still return a high set-level FDR (≥ 0.6) under both scans, while bio_winter’s floor test returns 10 of 18,361 units at $\text{FDR} \approx 0.18$ (Section 5) despite this section’s finding that bio_winter’s *overall* genome-wide BF landscape trends, if anything, *away* from high-DI and sorted regions – a handful of individually exceptional loci coexisting with a bulk distribution pulling in the opposite direction from what a simple ancestry-confound story would predict. None of these floor-survivor sets overlap physically: mapping each covariate’s floor-survivor units onto the canonical Stage-2 (rho05) clustering (eMLG_51loci_0025_cM05_rho05.rds) gives zero shared Stage-2 groups between any pair of PC1 (1 region), PC2 (2), mitoC2 (0), and bio_winter (7) – even the closest case, a PC1 and a bio_winter floor-survivor both on Chr24 only ~53 kb apart, falls into two distinct Stage-2 clusters (Figure 5).

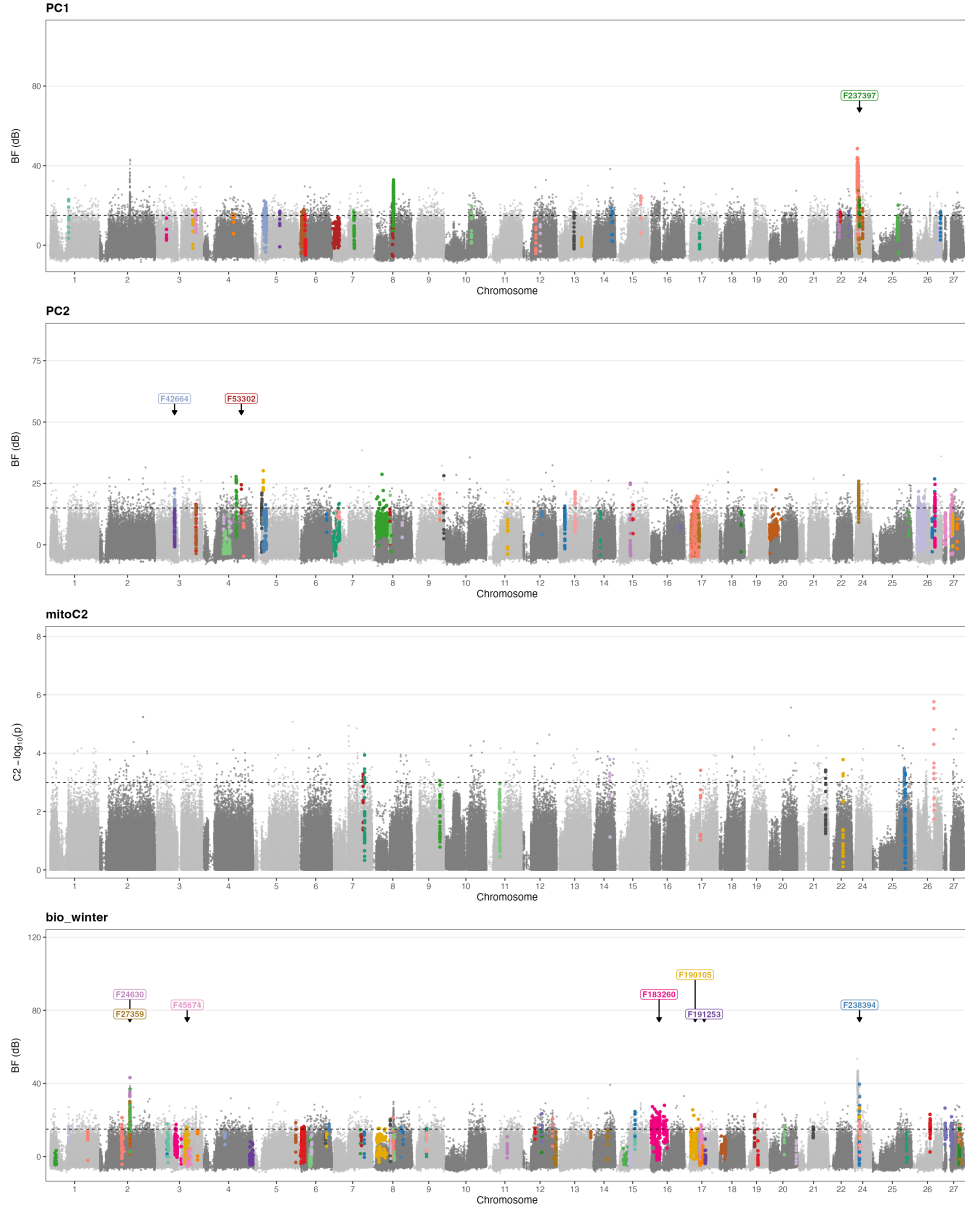


Figure 5: Stage-1-direct genome-wide scan, all four covariates. Every genome-wide SNP is plotted at its own full-SNP BayPass value – BF(dB) for PC1, PC2, and bio_winter (continuous-covariate regression); the C2 contrast statistic – $-\log_{10}(p)$ for mitoC2 (population contrast). Dashed line: the raw threshold (BF(dB) ≥ 15 , or $-\log_{10}(p) \geq 3$ for mitoC2). Grey (alternating shade by chromosome): no Stage-1 cluster in this canonical Stage-2 (rho05) region raw-crossed the threshold. Coloured (one colour per region, recycled): every Stage-2 (rho05) region containing ≥ 1 raw threshold crossing – most of these are raw crossings only, not validated candidates. Arrowed and labelled (Stage-2 group_id): the subset of coloured regions that additionally contain a floor-survivor Stage-1 unit (beats all 10,000 Ω -structured null draws), with the label coloured to match its region – PC1: 45 raw crossings $\rightarrow$ 34 regions coloured, 1 floor survivor $\rightarrow$ 1 region arrowed (F237397); PC2: 65 raw crossings $\rightarrow$ 42 regions coloured, 2 floor survivors $\rightarrow$ 2 regions arrowed (F42664, F53302); mitoC2: 11 raw crossings $\rightarrow$ 11 regions coloured, 0 floor survivors, no region arrowed; bio_winter: 113 raw crossings $\rightarrow$ 71 regions coloured, 10 floor survivors $\rightarrow$ 7 regions arrowed (F24630, F27359, F45674, F183260 [4 Stage-1 clusters], F190105, F191253, F238394). No arrowed region is shared between any pair of covariates (see text).

What Section 6 adds is a more granular picture: the Stage-1-direct climate-BF landscape is *not*

architecture-agnostic for any of the three continuous axes, but the departures point in different, and for sorting a uniformly unexpected, direction – so *which* architectural feature aligns with which axis, and whether that alignment reflects the same ancestry confound discussed in Sections 1–4 or a distinct mechanism, is not resolved here and would need its own follow-up (e.g. relating each axis’s DI/recombination/sorting correlation directly to that axis’s own ancestry-confound coefficient from `moduleB_ancestry_confound.R`).

Independent cross-check using BayPass’s own outlier-region method. A separate module (`module_localscore_crosscheck`, own report) re-examines the same four Stage-1-direct covariates with BayPass’s own recommended *local-score* outlier-region method (Fariello et al. 2017; Bonhomme et al. 2019), which builds candidate windows directly from each statistic’s spatial profile rather than from the Stage-2 LD-clustering used throughout this document. Run genome-wide (full-SNP resolution), it independently recovers PC1 and `bio_winter`’s shared architecture ($r = +0.89$ – 0.90 genome-wide, matching their near-collinear raw values and their near-identical significant windows), with PC2 – a genuinely different axis – showing neither. But a direct check of the method’s own significance threshold against the same Ω -structured nulls used throughout this pipeline shows it is not well calibrated for this dataset: roughly one in five structure-only null draws for the continuous covariates, and the clear majority for `mitoC2`’s contrast, already produce one or more “significant” windows by chance, and a single worst-of-20-random-draws null run through the actual full-SNP BayPass pipeline produced 104 convincing-looking significant windows from a covariate with zero true signal by construction – more than the real `mitoC2` covariate itself. Those null and real `mitoC2` windows also overlap more than chance predicts, traced to both ballooning specifically in the genome’s lowest-recombination, highest-LD regions (an artifact of the method’s window-growth mechanism, not of shared true signal). This independent method therefore reinforces the ancestry-confound reading of PC1/`bio_winter` above, but is not used as a source of candidate loci anywhere in this pipeline.

Limitations and recommended follow-up

This analysis should be treated as **supplementary and explicitly exploratory**, not a confirmed result, for several reasons:

- **Only PC1 received the full mitotype-controlled (partial) analysis.** PC2, bio6, and bio11 are examined only through bivariate correlations. The conclusion that no climate variable shows an ancestry- and mitotype-independent signal is broader than what has actually been tested for those three variables; either the same partial model should be run for all four, or the conclusion should be narrowed to PC1 specifically.
- **Ancestry was computed once, from the frozen $DI > -25$ marker panel.** It has not been recomputed from an independent marker set or from LD-reduced diagnostic units, so its sensitivity to the specific marker panel used is untested.
- **A single coherent structure-aware model is preferable to a sequence of bivariate/partial tests.** A model of the form $\text{ancestry} \sim \text{climate} + \text{mitotype}$ with an Ω -based residual covariance (or an equivalent parametric bootstrap) would estimate both predictors' contributions jointly, with correctly propagated uncertainty, rather than inferring their relative roles from separate bivariate/partial tests as done here.
- With 17 independent lineage units and 4-5 tests per family, this analysis is underpowered to detect anything but a fairly strong effect, and Ω -null and block-permutation p -values disagree by up to ~ 0.02 – 0.07 in places (e.g. PC2: 0.190 vs. 0.123) – a reminder that “significant” here would depend on which defensible null is treated as primary, not a settled question either null can resolve alone.
- **Section 6's genome-wide architecture calibration has mixed-direction results that were not chased down.** DI , recombination, and sorting depart from the Ω -structured null differently across PC1, PC2, and bio_winter, and for sorting in the opposite direction from a naive confounding argument on all three axes. Adding bio_winter and the mitoC2 calibration check (own dedicated null, no departure on any measure) strengthens confidence that these departures are not an artifact of the null-covariate machinery, but neither addition explains *why* the three continuous axes disagree on direction. Relating each axis's departure directly to that axis's own ancestry-confound coefficient (rather than to the coarser population-level ancestry summary used in Sections 1–4) would be needed before drawing a mechanistic conclusion from it.

Summary

Across the 19 Åland-excluded populations (17 independent lineage units after correcting for shared origins), genome-wide ancestry showed a moderate negative correlation with climate PC1 ($r = -0.48$). This relationship was nominally suggestive under the shared-lineage permutation ($p = 0.051$) but did not clear the Ω -structured null ($p = 0.074$, and the two nulls disagree), and it did not survive correction for testing four climate variables ($p_{BH} = 0.16$). The association was further attenuated and statistically indistinguishable from the population-structure null after controlling for mitotype ($r = -0.40$, $p = 0.12$ – 0.14). Mitotype was itself nominally associated with nuclear ancestry ($p \approx 0.05$) but not with any measured climate variable, and that association also did not survive multiple-testing correction. These patterns suggest contributions from population history and possibly climate, but the limited number of independent populations does not permit their effects to be separated confidently. Critically, this moderate, inconclusive population-level alignment is sufficient to explain the raw candidate counts in the original BayPass climate scan without invoking any locus-specific climate effect (Section 5): no PC1 cluster and only a null-expected number of PC2

clusters exceeded the Ω -structured null. The BayPass scan therefore provides no locus-specific evidence for climate adaptation beyond what population structure already predicts. A separate, genome-wide calibration (Section 6), extended here to `bio_winter` and to `mitoC2` as a specificity check, shows the Stage-1-direct climate-BF landscape is not architecture-agnostic for any of PC1, PC2, or `bio_winter` – seven of nine tests relating BF strength to Diagnostic Index, recombination, and directional sorting depart from a shared structured-null baseline – but the directions differ across the three axes and, for sorting, run counter to a simple confounding story on all three, so this result sharpens rather than resolves the population-level picture above. `mitoC2`, tested the same way against its own dedicated contrast-mode null, shows no such departure on any of the three measures ($p_{BH} = 0.963$ throughout), reinforcing that the PC1/PC2/`bio_winter` departures are a property of those covariates and not an artifact of the calibration framework itself. `bio_winter` (the combined, standardized `bio6/bio11` axis) remains the one climate result that stands apart from the rest of this ancestry-confound narrative at the individual-locus level: 10 of 18,361 Stage-1 units survive the floor test ($FDR \approx 0.18$, Section 5), clearly stronger than PC1 or PC2 achieve on either scan. Section 6 complicates rather than resolves that lead, however: `bio_winter`’s *overall* genome-wide BF distribution trends away from high-DI and directionally-sorted regions, the opposite of what would be expected if its floor-survivor loci reflected a broad ancestry-tracking signal – so those 10 units look more like a distinct, localized signal than the tip of a genome-wide ancestry-confound iceberg, though this is not established and remains an open lead for future work. This document should be read as a supplementary, exploratory analysis, not a confirmed finding for the main text.